



EVOPRIME

700

Constant-current LED power module

12–15 V

DC INPUT

700 mA

NOMINAL OUTPUT

6–15

SERIES LEDs

ENGINEERED BY GLIGHT

USER MANUAL · ENGLISH

VERSION 1.0 · FINAL ENGLISH EDITION

GLIGHT / GLIGHTCOREAURA · TOKYO, JAPAN

Contents

Installation, operation and diagnostic architecture.

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QUICK READING

For installation: 01 → 03 → 04 → 05/06 → 07 → 08. For a fault: 09.

EvoPrime operates with high output voltage. Keep this manual with the vehicle documentation.

Before applying power

Read these warnings before connecting the battery or power supply.

OUTPUT HAZARD LED+ can reach approximately 55 V with an open string. Fully disconnect power before touching, soldering or reconnecting the load.

CORRECT POLARITY REQUIRED EvoPrime 700 has no electronic reverse-battery protection. Reversing +12V and GND can destroy the module.

GL_SIG CONTROL GL_SIG is a low-side input for GLightOne. Never apply +12 V, battery voltage or a positive signal to this pad.

- Install the module only with power disconnected and the fuse removed.
- Use one series-connected power LED string only; do not connect parallel branches.
- Each LED must support 700 mA and have adequate thermal dissipation.
- Install the PCB inside the headlamp, secured on standoffs and protected from metal, condensation and conductive debris.
- Installation must be performed by a person experienced in automotive electrical systems and power LEDs.
- The installer is responsible for compliance with local vehicle-lighting regulations.

IMPORTANT The red PCB LED indicates FAULT. It is not a simple power indicator.

What EvoPrime 700 is

Power control for high-intensity LEDs within the GLight ecosystem.

EvoPrime 700 is a constant-current boost driver designed to power one string of power LEDs from a nominal 12 V system. Output is fixed at approximately 700 mA and requires no user adjustment.

It can operate independently or follow GLightOne switching and PWM through the low-side GL_SIG input and LINK_DET detection.

PARAMETER	OPERATING SPECIFICATION
Input	12–15 V DC; nominal 12 V system
Start-up	Approximately 11.1 V typical; not designed for 6 V systems
LED current	700 mA nominal
LED string	6–15 power LEDs, one series string only
Recommended total Vf	16–49 V at 700 mA
No-load voltage	Limited to approximately 55 V; do not touch while powered
Control	Standalone or low-side PWM from GLightOne
PCB	60 × 40 mm; 4 layers; 1.6 mm
Mounting	4 × Ø3.2 mm holes connected to GND

Integrated protections:

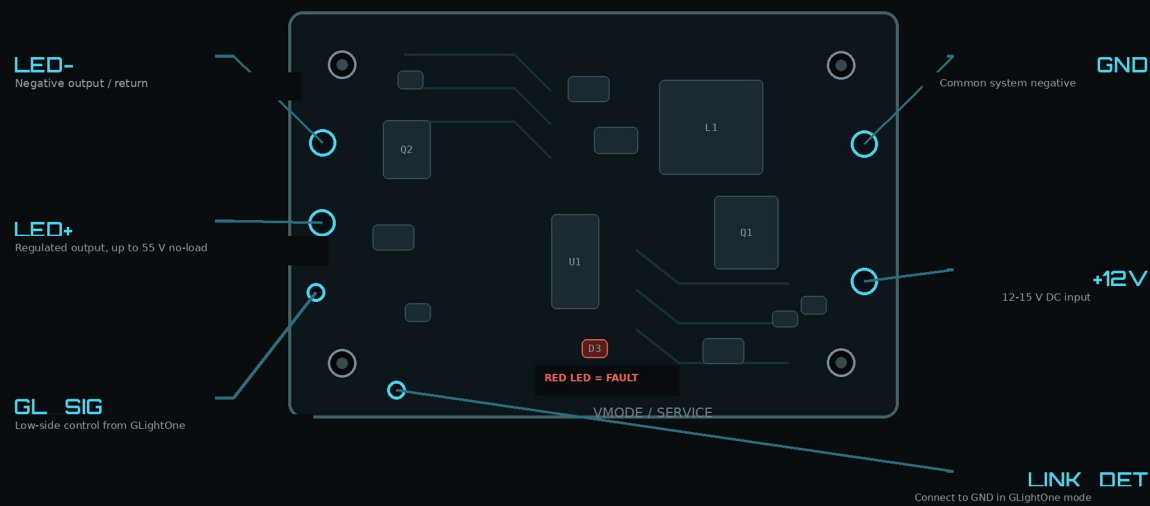
- Internal input fuse and transient suppression.
- Output disconnect, current limiting and short-circuit protection.
- Output overvoltage limiting and FAULT indication.
- Controller thermal protection.

KNOWN LIMITATION These protections do not replace correct polarity, an external fuse, insulation, ventilation or professional installation.

Connection identification

Top view of the final PCB. Orient the board using the printed labels.

EVOPRIME 700 / CONNECTION MAP



TOP VIEW / ACTUAL PCB ORIENTATION 60 × 40 mm

Map of user connections and reserved service points.

PAD	FUNCTION	RULE
+12V	Positive input	12–15 V DC; follow the arrow and polarity
GND	Common negative	Share with GLightOne when controlled
LED+	Regulated positive output	Connect to the string anode
LED-	LED return	Connect to the final string cathode
GL_SIG	Low-side control input	Never apply +12 V
LINK_DET	GLight mode detection	Connect to GND only in GLightOne mode

DO NOT CONNECT VMODE, ISMON, ISP and ISN are measurement/service points. They are not part of normal user wiring.

Selecting the LED string

LED count alone is not enough: the sum of their forward voltages must also be checked.

ONE STRING / ALWAYS IN SERIES



6-15 LEDs / RECOMMENDED TOTAL Vf 16-49 V @ 700 mA

DO NOT CONNECT PARALLEL BRANCHES

EvoPrime regulates one series string at 700 mA.

- 1. Confirm each LED's Vf at 700 mA and at the minimum expected temperature in its datasheet.
- 2. Add the maximum Vf of all LEDs in the string.
- 3. Use a total of approximately 16 V to 49 V.
- 4. Verify that every LED supports 700 mA continuously and has a heatsink.

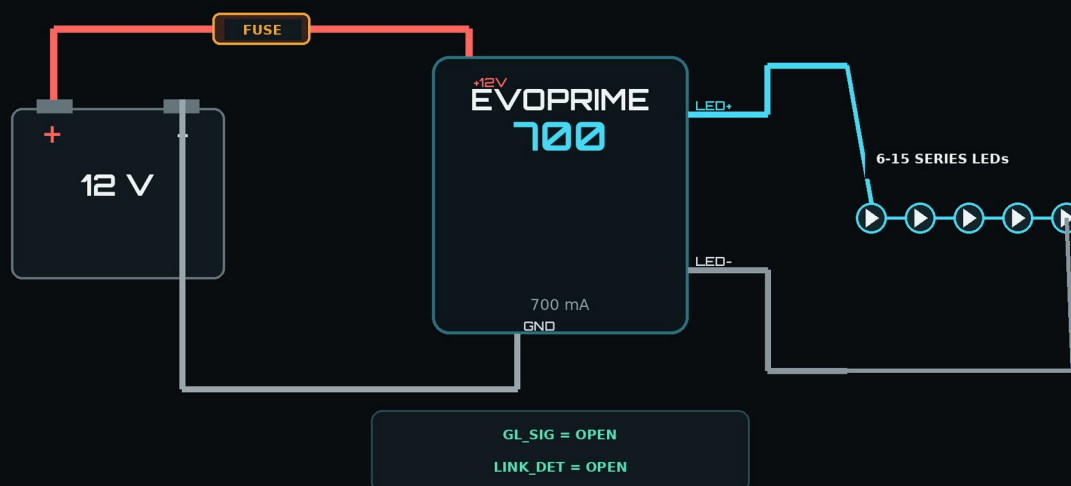
EXAMPLE	CALCULATION	RESULT
10 LEDs at 3.0 V	$10 \times 3.0 \text{ V} = 30 \text{ V}$	Correct
15 LEDs at 3.2 V	$15 \times 3.2 \text{ V} = 48 \text{ V}$	Correct, near the limit
15 LEDs at 3.5 V	$15 \times 3.5 \text{ V} = 52.5 \text{ V}$	Not recommended

DO NOT USE Bulbs, motors, resistors, constant-voltage LED strips, parallel LED branches or strings whose Vf exceeds the stated limit.

Standalone installation

EvoPrime turns on at full current when GL_SIG and LINK_DET are left open.

STANDALONE CONNECTION



Standalone wiring. Keep GL_SIG and LINK_DET disconnected and insulated.

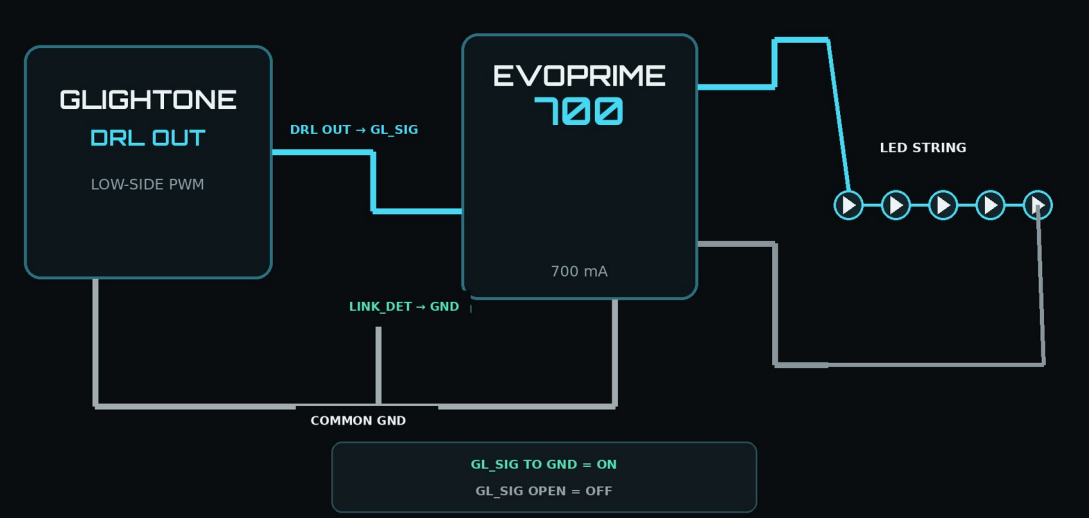
1. Remove the external fuse or fully disconnect the supply.
2. Connect the series string: first anode to LED+ and final cathode to LED-.
3. Connect GND to the negative terminal of the supply or battery.
4. Connect +12V to the positive terminal through an external fuse near the source.
5. Leave GL_SIG and LINK_DET open, separated and insulated.
6. Check polarity, insulation and heatsinks before reinstalling the fuse.

BEHAVIOR Above the start-up threshold, the string turns on at approximately 700 mA. Remove power to turn it off.

Control with GLightOne

Direct integration with GLightOne's low-side DRL OUT.

CONTROL FROM GLIGHTONE



GLightOne and EvoPrime must share GND. LINK_DET selects controlled mode.

1. Disconnect power from both modules.
2. Join the GND connections of GLightOne, EvoPrime and the supply.
3. Connect GLightOne DRL OUT to the EvoPrime GL_SIG pad.
4. Connect EvoPrime LINK_DET to GND.
5. Connect +12V, LED+ and LED- as described in the previous sections.

GL_SIG STATE	EVOPRIME RESULT
GL_SIG connected to GND by GLightOne	On / full current
GL_SIG released by GLightOne	Off
GL_SIG with low-side PWM	Brightness follows PWM duty cycle
LINK_DET without GND	Standalone mode; remains on

CRITICAL GL_SIG accepts low-side control only; never +12 V.

Mounting, wiring and environment

Reliability depends on the installation as much as on the circuit.

MOUNTING DIMENSIONS



Nominal PCB dimensions and mounting-hole pattern.

- Secure the board with standoffs; do not allow the underside to touch metal or hardware at any potential other than GND.
- All four mounting holes are electrically connected to GND.
- Install inside the headlamp in an area protected from water and condensation, away from direct heat from the projector or optics.
- Leave clearance around L1, Q1, D4 and the current-sense resistors. None of these components may touch acrylic, the lens, reflector or headlamp plastic.
- Use automotive wire, secure terminals and strain relief. Prevent wiring from pulling on GL_SIG or LINK_DET.
- Place the external fuse as close as possible to the +12 V source and size it for the wiring and final load.

WIRE RECOMMENDATION For short runs: minimum 0.75 mm² (AWG 18) input and minimum 0.5 mm² (AWG 20) LED output. Increase conductor size for long runs.

Initial start-up

Perform this check after every installation or LED-string change.

Inspection with power disconnected:

- +12V and GND match the PCB arrows.
- LED+ reaches the first anode and LED- the final cathode.
- All LEDs are in series and mounted on heatsinks.
- The total V_f is within 16–49 V.
- There are no loose strands, solder bridges or chassis contact.
- GL_SIG/LINK_DET match the selected mode.
- The external fuse is installed near the source.

Power-up sequence:

1. If a bench supply is available, initially limit input current and raise the voltage gradually to 12 V.
2. Apply power for a few seconds and confirm there is no odor, clicking, restart, FAULT or immediate heating. A faint stable tone during PWM may be normal; see the diagnostic table.
3. Check on, off and PWM behavior for the selected mode.
4. Keep the system powered and check the temperature of the board, connectors, wiring and LED heatsinks.
5. Disconnect power before correcting any connection.

NEVER HOT-CONNECT Do not connect or disconnect the LED string while EvoPrime is powered. The output may retain or reach high voltage.

TONE DURING PWM A faint stable tone may occur due to mechanical vibration of L1, C18 or the mounting. By itself it does not indicate a fault if the light remains stable, FAULT is off and there is no abnormal heating.

STOP if there is clicking, restarting, irregular flicker, persistent FAULT, odor or rapid heating.

Basic diagnostics

The red LED indicates a FAULT condition; its absence does not mean the output is safe to touch.

SYMPTOM	PROBABLE CAUSE	ACTION
Does not turn on; red LED off	No 12 V, polarity/wiring fault, voltage below start-up, or GLight mode active without GL_SIG at GND	Disconnect power; measure input and check LINK_DET/GL_SIG
Red LED on	Overvoltage/open output, overcurrent, short circuit or temperature	Disconnect power; check string, polarity, cooling and shorts
Always on with GLightOne	LINK_DET is not connected to GND	Connect LINK_DET to common GND
Never turns on with GLightOne	GL_SIG receives no pull-down or there is no common GND	Check low-side DRL OUT and GND continuity
Flicker or unstable brightness	Weak supply, poor GND, long wiring or incompatible PWM	Check voltage drop, terminals and signal
Turns off when hot	Insufficient cooling or excessive ambient temperature	Improve ventilation; check load and mounting
Faint tone during PWM	Mechanical vibration of L1/C18 or mounting resonance	See section 09; stop if FAULT, flicker, restart or abnormal heat occurs

Recovery after FAULT:

1. Disconnect +12 V completely.
2. Wait for the board and LEDs to cool.
3. Correct the cause before applying power again.
4. Do not repeat power cycles if the red LED reappears.

STOP If there is visible damage, odor, charring, a fractured component or an open fuse, do not reconnect the module.

Quick reference

Installer summary. Keep this page with the vehicle or project.

CONNECTION	DESTINATION
+12V	12–15 V DC through an external fuse
GND	Common negative
LED+	First anode of the LED string
LED-	Final cathode of the LED string
GL_SIG	GLightOne low-side DRL OUT; never +12 V
LINK_DET	To GND in GLightOne mode; open in standalone mode

MODE	GL_SIG	LINK_DET	RESULT
Standalone	Open	Open	On when powered
GLightOne ON	To GND	To GND	On / PWM
GLightOne OFF	Open	To GND	Off

FIVE RULES 12–15 V only · correct polarity required · one series LED string · total Vf 16–49 V · never connect or disconnect the string while powered.

Support and documentation:

GLIGHTCORE.COM

This manual applies to EvoPrime 700. Do not use these connections with another revision or variant without confirming its specific documentation.

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ENGINEERED BY GLIGHT · TOKYO, JAPAN